

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		potassium / K^+ bromide / Br^- do not accept bromine calcium / Ca^{2+} iodide / I^- do not accept iodine award (3) for all four ions correct award (2) for any two correct award (1) for any one correct			3	3		3
		(ii)		$BaCl_2$		1		1		
	(b)			$Na^+(aq) + NO_3^-(aq) \longrightarrow AgCl(s)$ ☐ $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s) + AgCl(s)$ ☐ $Ag^+(aq) + NO_3^-(aq) \longrightarrow AgNO_3(s)$ ☐ $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$ ☒ $Ag^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s)$ ☐		1		1		

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	(c)			from the equation 552 g of Ag ₂ CO ₃ gives 432 g of Ag (1) 1 g of Ag ₂ CO ₃ gives $\frac{432}{552}$ g of Ag (1) 13.8 g of Ag ₂ CO ₃ $\frac{432}{552} \times 13.8$ g of Ag = 10.8 g (1) ecf possible alternative method moles of Ag ₂ CO ₃ = $\frac{13.8}{276} = 0.05$ mol (1) 2 mol of of Ag ₂ CO ₃ gives 4 mol of Ag therefore 0.10 mol of Ag produced (1) mass of Ag = n × M _r = 0.10 × 108 = 10.8 g (1) ecf possible second alternative method (applying conservation of mass) percentage of Ag in Ag ₂ CO ₃ = $\frac{216}{276} \times 100 = 78.3\%$ (1) mass of Ag on left-hand side = $\frac{78.3}{100} \times 13.8 = 10.8$ g (1) so 10.8 g of Ag formed (1) ecf possible						
				Question 9 total	0	5	3	8	3	3